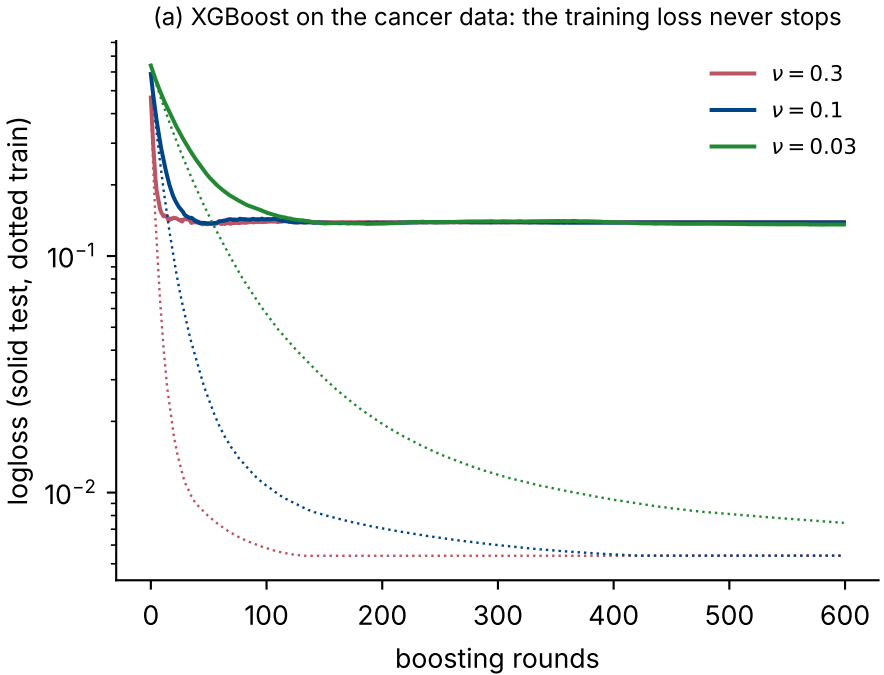
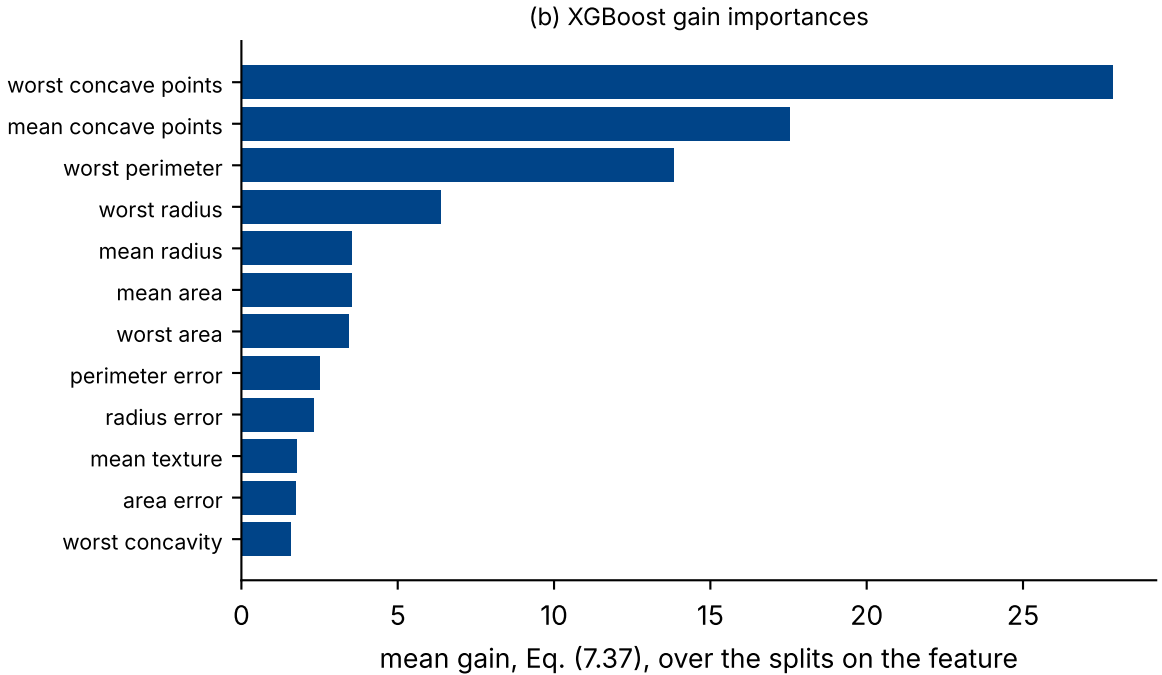

(a) XGBoost on the cancer data: the training loss never stops
$\nu = 0.3$
$\nu = 0.1$
$\nu = 0.03$
logloss (solid test, dotted train)
10^{-1}
10^{-2}
0
100
200
300
400
500
600
boosting rounds


(b) XGBoost gain importances
worst concave points
mean concave points
worst perimeter
worst radius
mean radius
mean area
worst area
perimeter error
radius error
mean texture
area error
worst concavity
0
5
10
15
20
25
mean gain, Eq. (7.37), over the splits on the feature